# SACHEM

Cleburne, TX

***Report No.:*** 54-11-001-11
***Inspector :*** Steve Sherman
***Employer :*** OilPro Consulting LLC
***Inspection Date :*** 06/20/2017

***IN-SERVICE***

***Inspection Report For***

# Pressure Vessel

***54-11-001***

An API Standard 510 Inspection based on client criterion for nondestructive examinations was conducted on vessel 54-11-001 in the SACHEM facility located at Cleburne, TX on 06/20/2017. This vessel was originally built to ASME S8 D1. This inspection was conducted in accordance with requirements of the API-510 standard for inspections of Pressure Vessels. The following is a detailed report of the inspection including findings and recommendations.

***Inspector Signature***
***API Certification No.***

***JERRY HARTFIELD***
***MGR.***

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# *1.0 EXECUTIVE SUMMARY*

An API Standard 510 In Lieu Of Internal inspection was preformed on pressure vessel 54-11-001 located at CLEBURNE, TX and was conducted on 2/24/2017. This inspection was made to collect data in order to evaluate the mechanical integrity and fitness for service of the vessel. This inspection consisted of an External VT, UT examination on the heads and the Shell of the vessel and an Ultrasonic Scan of the Lower portion of the vessel with no major problems noted during this inspection with minor discrepancies listed in Section 3.0 Inspection Results and 4.0 Recommendations.

4.8.1 Next external inspection by SACHEM in-house representatives is due by: 06/20/2018

4.8.1 Next external inspection by an API 510 Inspector is due by: 06/20/2022

4.8.2 Next internal inspection is due by: 06/20/2027

4.8.3 Next UT inspection is due by: 06/20/2018

4.8.4 Governing component limiting life: Shell 1

***TABLE A***

| *Component* | *Nominal Design Thickness (in.)* | *Actual Measured Thickness (in.)* | *Minimum Required Thickness (in.)* | *Design MAWP (psi) Internal* | *Calculated MAWP (psi) Internal* | *Remaining Life (years)* |
|---|---|---|---|---|---|---|
| Shell 1 | 0.813 | 0.796 | 0.719 | 225 | 246.3 | 180 |
| North Head | 0.530 | 0.497 | 0.421 | 225 | 257.1 | 92.1 |
| South Head | 0.530 | 0.493 | 0.421 | 225 | 253.8 | 77.9 |

**Next Inspection:**

Next external inspection is due by: 6/20/2022
Next internal inspection is due by: 6/20/2027
Next UT inspection is due by: 6/20/2018

# *2.0 VESSEL DATA*

| ***Vessel Data:*** | | ***General Data:*** | |
|---|---|---|---|
| ***MAWP (psi):*** | 225 | ***Product:*** | TriMethylamine |
| ***Design Temp. (degF):*** | 125 | ***Year Built:*** | 1977 |
| ***MDMT (degF):*** | -20 | ***NB No.:*** | 51109 |
| ***Oper. Press.(psi):*** | 175 | ***Const Code*** | ASME S8 D1 |
| ***Oper. Temp. (degF):*** | | ***Vessel Config.:*** | Horizontal |
| ***Material Type:*** | Carbon Steel | ***Head Type:*** | Hemispherical |
| ***Inside Dia. (inch):*** | 130.250 | ***Insul. Type:*** | None |
| ***Length (inch)*** | 432.500 | ***Insul. Thk (inch):*** | 0 |

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| | |
|---|---|
| ***MAWP (psi):*** | ***Oper. Press.(psi):*** |
| ***Design. Temp. (degF):*** | ***Oper. Temp.(degF):*** |
| ***Material:*** | ***Medium:*** |

# *3.0 INSPECTION RESULTS, IN-SERVICE*

***The following results are the summarization of a field checklist that was utilized during the inspection of vessel 54-11-001***

## *3.1 Foundation:*

3.1.1 The vessel has 2 carbon steel foundation supports attached to the lower section of the horizontal Storage Tank.

3.1.2 The supports are in satisfactory mechanical condition. The supports coating is in satisfactory condition.

NOTE 1: Possible corrosion on the exteriror shell surface of the vessel at the support to shell contact area, that was notated in February 11, 2016 inspection report was re-inspected.
•UT thickness readings were performed on 100% of the accessible surfaces of the vessel's supports to detect any potential areas of concern.
•VT of 100% Of the perimeter of the vessel's supports shell to detect blistering, discontinuity in shape, scaling or rust stains beneath the supports

## *3.2 Shell:*

3.2.1 The shell is un-insulated, carbon steel with 20 to 30 mils epoxy external coating

3.2.2 The external surface profile of the shell appears to be smooth and clean and in satisfactory condition.

3.2.3 The exposed surface profile of the shell is relatively smooth with no evidence of significant oxidation or other forms of deterioration.

3.2.4 The longitudinal and circumferential welds of the shell appeared to be in
satisfactory condition with no preferential corrosion or indications of cracking observed.

Note 2: Phased Array was performed on all T-junctions and 30% of the vessel welds in the HAZ "Heat affected zone". Areas with the Highest calculated stresses were chosen for the makeup of the 30%.

3.2.5 The shell nozzle penetration welds appeared to be in satisfactory condition.

Note 3: Phased Array was performed on the nozzles in a 6 inch radius from the welds
to inspect for potential cracking and any other discontinuities. No significant indications were recorded.

Note 4: An ultrasonic thickness scan was performed on the lower portion of the vessel on grids spaced at 6'' apart traveling from north to south on the lower 40% of the vessel shell to look for potential corrosion caused by the possibility of moisture getting into the system. The decision to scan the lower part of the vessel was made based on our conversations with SACHEM's internal Chemical Engineer. He explained that if any moisture were to get into the system it would mothball and fall to the bottom of the vessel. The write up is attached as one of the addendum at the end of this report.

## *3.3 Head(s):*

3.3.1 The north and south heads are torispherical in design, un-insulated, carbon steel, with 20 to 30 mils of epoxy external coating.

3.3.2 The external surface profiles of both heads have a smooth finish. There are no areas of coating failure.

3.3.3 Exposed surface profiles were relatively smooth with no evidence of significant oxidation or other forms of deterioration.

3.3.4 The nozzle penetration welds through the top and bottom heads appeared to be in satisfactory condition.

Note 5: An ultrasonic thickness measurements were taken on a grid by NIS in addition to the CML's that are monitored yearly. the report is included as an addendum to this report found at the end.

### *3.4 Appurtenances:*

3.4.1 The shell and head-nozzles appear to be clean and in satisfactory condition.

3.4.2 The vessel has a inch Pressure Safety Relief Valve (PSV) set at 225 psi and is thereby protected from over pressure. The PSV was certified on and appears to be in compliance with code requirements.

3.4.3

3.4.4 The ASME name-plate is attached, readable and is easily accessable

# *4.0 RECOMMENDATIONS*

## *4.1 Foundation:*

4.1 Foundation:
4.1.1 No Recommendations

## *4.2 Shell:*

4.2.1 An in-lieu-of internal inspection was granted with the following stipulations:
•UT's must be performed annually for 10 years or until an internal inspection is performed
•OilPro Consulting must review the UT's and assess the vessel for continued s

## *4.3 Heads:*

4.3.1 1 An in-lieu-of internal inspection was granted with the following stipulations for the heads:
•UT's must be performed annually for 10 years or until an internal inspection is performed
•OilPro Consulting must review the UT's and assess the heads

## *4.4 Appurtenances:*

4.4.1 An in-lieu-of internal inspection was granted with the following stipulations for the appurtenances:
•UT's must be performed annually for 10 years or until an internal inspection is performed
•OilPro Consulting must review the UT's and assess the

## *4.5 Next Inspections:*

4.8.1 Next external inspection is due by: 06/20/2022

4.8.2 Next internal inspection is due by: 06/20/2027

4.8.3 Next UT inspection is due by: 06/20/2018

4.8.4 Governing component limiting life: South Head

# 5.0 ULTRASONIC THICKNESS (UT) MEASUREMENTS

## 5.1 Results Summary:

5.1.1 UT measurement of accessible vessel components (shell, heads and nozzles) found no significant material loss due to internal corrosion of the components. All of the vessel component thicknesses were above that required by ASME calculations for minimum required thicknesses for internal pressure.

5.1.2 Calculations of all evaluated components resulted in greater than 20 years remaining life.

## 5.2 Recommendations:

5.2.1 External inspections and UT's of the vessel shall be scheduled annually for 10 years in accordance with the current In-Lieu-Of Internal Inspection granted in this report. The UT's and external inspections along with photos must be rev

# APPENDIX A

## Thickness Measurement Record

1) Component Thickness Measurements

# APPENDIX B

## Mechanical Integrity Calculation

1) Report Cylindrical Shell Calculations
2) Report Formed Head Calculations
3) Report Nozzle Calculation
4) Horizontal Tank Evaluation

# API-510 PRESSURE VESSEL SHELL EVALUATION

**MINIMUM THICKNESS, REMAINING LIFE, PRESSURE CALCULATIONS**

| Report No. | Client | Inspector | Vessel | Date |
|---|---|---|---|---|
| 54-11-001-11 | SACHEM | Steve Sherman | 54-11-001 | 6/20/2017 |

| Shell 1 | | | Material | | | | | | Material | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | CS - A612-A | | | | | | | | |
| Temp. | MAWP | SH | SG | D | t nom | Temp. | MAWP | SH | SG | D | t nom |
| 125 | 225 | 8.0 | 0.63 | 130.250 | 0.813 | | | | | | |

## Minimum Thickness Calculations

| Shell 1 | | Internal | PR/(SE-0.6P) = t | | | | Internal | PR/(SE-0.6P) = t | |
|---|---|---|---|---|---|---|---|---|---|
| P | R | S | E | t | P | R | S | E | t |
| 227.2 | 65.125 | 20700 | 1 | 0.719 | | | 0 | | |

## Remaining Life Calculations

| Shell 1 | t prev | t act | t min | y | | t prev | t act | t min | y |
|---|---|---|---|---|---|---|---|---|---|
| | 0.813 | 0.796 | 0.719 | 40.0 | | | | | |

| | | | | | | |
|---|---|---|---|---|---|---|
| Ca = t act - t min | = | 0.077 | (inch) | Ca = t act - t min | = | (inch) |
| Cr = t prev - t act / Y | = | 0.00042 | (in/year) | Cr = t prev - t act / Y | = | (in/year) |
| RL= Ca / Cr | = | 180.0 | (year) | RL= Ca / Cr | = | (year) |

**Next Inspection (Yn) =** 10 (years)

## MAWP Calculations

| Shell 1 | MAP - Next Inspection | | | MAP - Next Inspection | | |
|---|---|---|---|---|---|---|
| Where t = t act - 2YnCr = | | 0.788 | (inch) | Where t = t act - 2YnCr = | | (inch) |
| SEt/R+0.6t = P = | | 248.5 | (psi) | SEt/(R+0.6t) = P = | | (psi) |
| P-(SH*.433*SG) = MAWP = | | 246.3 | (psi) | P-(SH*.433*SG) = MAWP = | | (psi) |

| API-510 PRESSURE VESSEL SHELL EVALUATION | | | | |
|---|---|---|---|---|
| MINIMUM THICKNESS, REMAINING LIFE, PRESSURE CALCULATIONS | | | | |
| ***Report No.*** | ***Client*** | ***Inspector*** | ***Vessel*** | ***Date*** |
| 54-11-001-11 | SACHEM | Steve Sherman | 54-11-001 | 6/20/2017 |

***Variable Definitions for Shell Calculations:***

*A = factor determined from Fig. G in Subpart 3 of Section II, Part D and used to enter the applicable material chart in Subpart 3 of Section II, Part D. For the case of cylinders having Do /t values less than 10, see UG-28(c)(2).*

*B = factor determined from the applicable material chart in Subpart 3 of Section II, Part D for maximum design metal temperature, psi [see UG-20(c)]*

*Ca = remaining corrosion allowance of the vessel part under consideration, in inches.*

*Cr = corrosion rate of the vessel part under consideration, in inches per year.*

*D = inside diameter of the shell course under consideration, in inches.*

*E = (Internal Calculations) lowest efficiency of any joint in the shell course under consideration. For welded vessels, use the efficiency specified in UW-12.*

*E = (External Calculations) - Modulus of Elasticity (MOE) at operating temperature for specified material,*
*P = the design maximum allowable internal working pressure, including static head pressure, in psi.*

*Pa = maximum allowable external working pressure (includes jacket pressure and vessel internal negative pressure) in psi.*

*R = inside radius of the shell under consideration, in inches.*

*Ro = outside radius of the shell under consideration, in inches.*

*RL = estimated remaining life of the vessel part under consideration, in years.*

*S = maximum allowable stress value, in psi.*

*SH = static head, in feet*

*SG = specific gravity of vessel product.*
*t = thickness of the vessel part under consideration, variable related to applicable calculation used therein, in inches.*

*t act = actual thickness measurement of the vessel part under consideration, as recorded at the time of inspection, in inches.*

*t min = minimum required thickness of vessel part under consideration, as the nominal thickness minus*

*the design corrosion allowance or the calculated minimum required thickness at the design MAWP at the coinciding working temperature, in inches.*

*t nom = design nominal thickness of head, in inches.*
*t prev = previous thickness measurement of the vessel part under consideration, as recorded at last inspection or nominal thickness if no previous thickness measurements, in inches.*

*Y = time span between thickness readings or age of the vessel if t nom is used for t prev, in years*

*Yn = estimated time span to next inspection of the vessel part under consideration, in years.*

# API-510 PRESSURE VESSEL HEAD EVALUATION

**MINIMUM THICKNESS, REMAINING LIFE, PRESSURE CALCULATIONS**

| Report No. | Client | Inspector | Vessel | Date |
|---|---|---|---|---|
| 54-11-001-11 | SACHEM | Steve Sherman | 54-11-001 | 6/20/2017 |

## *Vessel Head(s)*

**North Head and South Head**

| *MAWP* | *D* | *T* | *E* | *SG1* | *SG2* |
|---|---|---|---|---|---|
| 225 | 130.250 | 125 | 0.85 | 0.63 | 0.63 |

| *MAWP* | *D* | *T* | *SH* | *SG3* | *E* |
|---|---|---|---|---|---|
| | | | | | |

| *Head ID* | *Head Type* | *t nom* |
|---|---|---|
| North Head | Hemispherical | 0.530 |

| *Material* | *S* | *SH* | *P* |
|---|---|---|---|
| CS - A612-A | 20700 | 8.0 | 227.2 |

| *Head ID* | *Head Type* | *t nom* |
|---|---|---|
| South Head | Hemispherical | 0.530 |

| *Material* | *S* | *SH* | *P* |
|---|---|---|---|
| CS - A612-A | 20700 | 8.0 | 227.2 |

| *Head ID* | *Head Type* | *t nom* |
|---|---|---|
| | | |

| *Material* | *S* | *P* |
|---|---|---|
| | | |

***For Torispherical Heads***

| *L =* | *R =* | *M =* | *L =* | *R =* | *M =* | *L =* | *R =* | *M =* |
|---|---|---|---|---|---|---|---|---|
| | | | | | | | | |

## *Minimum Thickness Calculations*

| *Internal;* | *Hemispherical Head* | *2:1 Ellipsodial Head* | *Torispherical Head* |
|---|---|---|---|

***PL/(2SE-0.2P) = t min PD/(2SE-0.2P) = t min (knl) PLM/(2SE-0.2P) = t min (knl)***

***P0.9D/(2S-0.2P) = t min (crwn) PL/(2S-0.2P) = t min (crwn)***

***(reference supplemental calcs for other head type formulas)***

| | | |
|---|---|---|
| ***North Head*** | ***Hemispherical*** | ***t min =*** *0.421 (inch)* |
| ***South Head*** | ***Hemispherical*** | ***t min =*** *0.421 (inch)* |
| | | ***t min =*** *(inch)* |

# API-510 PRESSURE VESSEL HEAD EVALUATION

**MINIMUM THICKNESS, REMAINING LIFE, PRESSURE CALCULATIONS**

| Report No. | Client | Inspector | Vessel | Date |
|---|---|---|---|---|
| 54-11-001-11 | SACHEM | Steve Sherman | 54-11-001 | 6/20/2017 |

## *Remaining Life Calculations*

| *North Head* | *t prev* | *t act* | *t min* | *y* |
|---|---|---|---|---|
| | 0.530 | 0.497 | 0.421 | 40.0 |

| | | | |
|---|---|---|---|
| *Ca = t act - t min* | = | 0.076 | (inch) |
| *Cr = t prev - t act / Y* | = | 0.00083 | (in/year) |
| *RL= Ca / Cr* | = | 92.1 | (years) |

| | *t prev* | *t act* | *t min* | *y* |
|---|---|---|---|---|
| | | | | |

| | | | |
|---|---|---|---|
| *Ca = t act - t min* | = | | (inch) |
| *Cr = t prev - t act / Y* | = | | (in/year) |
| RL= Ca / Cr | = | | (years) |

| *South Head* | *t prev* | *t act* | *t min* | *y* |
|---|---|---|---|---|
| | 0.530 | 0.493 | 0.421 | 40.0 |

| | | | |
|---|---|---|---|
| *Ca = t act - t min* | = | 0.072 | (inch) |
| *Cr = t prev - t act / Y* | = | 0.00093 | (in/year) |
| *RL= Ca / Cr* | = | 77.9 | (years) |

***Next Inspection (Yn)*** 10 (year)

## *MAWP Calculations* *(reference supplemental calcs for other head type formulas)*

| *MAP* | *Hemispherical Head* | *2:1 Ellipsodial Head* | *Torispherical Head* |
|---|---|---|---|
| *Internal* | *2SEt/(R+0.2t) = P* | *2SEt/(D+0.2t) = P (knl)* | *2SEt/(LM+0.2t) = P (knl)* |
| | | *2St/(0.9D+0.2t) = P (crwn)* | *2St/(L+0.2t) = P (crwn)* |

***Where t = t act - 2YnCr*** ***Where P = MAP at the Next Inspection***

***Where MAWP = P-(SH*.433*SG)***

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| *North Head* | *Hemispherical* | *t =* 0.481 | *(inch)* | *P=* 259.3 | *(psi)* | *MAWP =* 257.1 | *(psi)* |
| *South Head* | *Hemispherical* | *t =* 0.475 | *(inch)* | *P=* 256.0 | *(psi)* | *MAWP =* 253.8 | *(psi)* |
| | | *t =* | *(inch)* | *P=* | *(psi)* | *MAWP =* | *(psi)* |

| API-510 PRESSURE VESSEL SHELL EVALUATION | | | | |
|---|---|---|---|---|
| MINIMUM THICKNESS, REMAINING LIFE, PRESSURE CALCULATIONS | | | | |
| Report No. | Client | Inspector | Vessel | Date |
| 54-11-001-11 | SACHEM | Steve Sherman | 54-11-001 | 6/20/2017 |

***Variable Definitions for Shell Calculations:***

*A = factor determined from Fig. G in Subpart 3 of Section II, Part D and used to enter the applicable material*

*chart in Subpart 3 of Section II, Part D. For the case of cylinders having Do /t values less than 10, see UG-28(c)(2).*

*B = factor determined from the applicable material chart in Subpart 3 of Section II, Part D for maximum design metal*

*temperature, psi [see UG-20(c)]*
*Ca = remaining corrosion allowance of the vessel part under consideration, in inches.*

*Cr = corrosion rate of the vessel part under consideration, in inches per year.*

*D = inside diameter for the head skirt, in inches.*

*Do = outside diameter for the head skirt, in inches.*

*E = (Internal Calculations) lowest efficiency of any joint in the vessel part under consideration. For welded vessels, use the efficiency specified in UW-12.*

*E = (External Calculations) - Modulus of Elasticity (MOE) at operating temperature for specified material.*

*h = inside height for the head, in inches.*

*ho = outside height for the head, in inches.*

*H = inside head height from tangent, in inches.*

*K = factor depending on head proportions D/2h and determined as 1/6[2+(D/2h)^2]*

*Ko = factor depending on head proportions Do/2ho (see table UG 33.1)*
*L = Hemi. Hds - inside spherical or crown radius of the head under consideration, in inches,. Ellip. and Tor.*

*Hds - inside spherical or crown radius, in inches,*

*M = a factor used in the formulas for torispherical heads where M= 0.25(3+(L/r)^.5)*

*P = the design maximum allowable internal working pressure, including static head pressure, in psi.*

*Pa = maximum allowable external working pressure, including static head pressure, in psi.*

*r = inside knuckle radius, in inches.*

*R = inside radius of the shell under consideration, in inches.*

*Ro = Hemi - outside radius in inches. Ellip. - the equivalent outside spherical radius taken as KoDo.*

*Tor. - outside radius of the crown portion of the head, in inches.*

| API-510 PRESSURE VESSEL SHELL EVALUATION | | | | |
|---|---|---|---|---|
| MINIMUM THICKNESS, REMAINING LIFE, PRESSURE CALCULATIONS | | | | |
| **Report No.** | **Client** | **Inspector** | **Vessel** | **Date** |
| **54-11-001-11** | **SACHEM** | **Steve Sherman** | **54-11-001** | **6/20/2017** |

*RL = estimated remaining life of the vessel part under consideration, in years.*

*S = maximum allowable stress value, in psi.*

*SH = static head, in feet*

*SG = specific gravity of vessel product.*

*t = thickness of the vessel part under consideration, variable related to applicable calculation used therein, in inches.*

*t act = actual thickness measurement of the vessel part under consideration, as recorded at the time of inspection, in inches.*

*t min = minimum required thickness of vessel part under consideration, as the nominal thickness minus the design*
*corrosion allowance or the calculated minimum required thickness at the design MAWP at the coinciding working*
*temperature, in inches.*

*t nom = design nominal thickness of head, in inches.*

*t prev = previous thickness measurement of the vessel part under consideration, as recorded at last inspection or*
*nominal thickness if no previous thickness measurements, in inches.*

*Y = time span between thickness readings or age of the vessel if t nom is used for t prev, in years.*

*Yn = estimated time span to next inspection of the vessel part under consideration, in years.*

**API-510 PRESSURE VESSEL NOZZLE EVALUATION**

***MINIMUM THICKNESS, REMAINING LIFE, PRESSURE CALCULATIONS***

| Report No | Client | Inspector | Vessel | Date |
|---|---|---|---|---|
| 54-11-001-11 | SACHEM | Steve Sherman | 54-11-001 | 06/20/2017 |

***Minimum Thickness Determinations:***

***a) The following nozzle minimum thicknesses are based on current ASME Standards. Minimum thickness allowed for nozzles walls are based on standard pipe thicknesses minus 12.5% or connecting shell/head required thickness whichever is smaller. (ASME Sect VIII, UG-45)***

| Size (inch) | 3/4 | 1 | 1.5 | 2 | 3 | 4 | 6 | 8 | 10 | 12 | >12 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| tmin (inch) | 0.100 | 0.116 | 0.127 | 0.135 | 0.189 | 0.207 | 0.245 | 0.282 | 0.319 | 0.328 | 0.328 |

***b) Large size nozzles or nozzles subject to high pressures are calulated per ASME Sect VIII, DIV 1, UG-27 as follows: PR/(SE-0.6P) = t.***

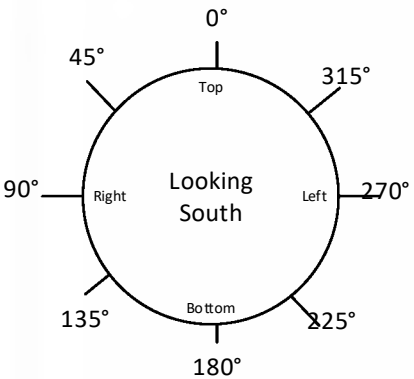
**Nozzle Remaining Life Calculations:**

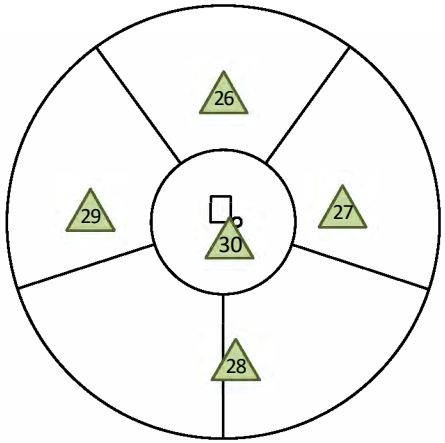
| CML | Noz ID. | Size (inch) | Material | Age (year) | t prev (inch) | t act (inch) | t min (inch) | Ca (inch) | Cr (inch/Yr) | RL (year) |
|---|---|---|---|---|---|---|---|---|---|---|
| 171 | N1 | 18 | CS - A516 70 | 40.0 | 0.562 | 0.575 | 0.328 | 0.247 | 0 | >20 |
| 172 | N2 | 2 | CS - A516 70 | 40.0 | 0.154 | 0.162 | 0.135 | 0.027 | 0 | >20 |
| 173 | N3 | 1 | CS - A106 B | 40.0 | 0.179 | 0.163 | 0.116 | 0.047 | 0.0004 | 117.5 |
| 174 | N4 | 2 | CS - A516 70 | 40.0 | 0.218 | 0.214 | 0.135 | 0.079 | 0.0001 | 790 |
| 175 | N5 | 3 | CS - A516 70 | 40.0 | 0.300 | 0.3 | 0.189 | 0.111 | 0 | >20 |
| 176 | N6 | 3 | CS - A516 70 | 40.0 | 0.300 | 0.302 | 0.189 | 0.113 | 0 | >20 |
| 177 | N7 | 1 | CS - A105 | 40.0 | 0.133 | 0.165 | 0.116 | 0.049 | 0 | >20 |

# APPENDIX C

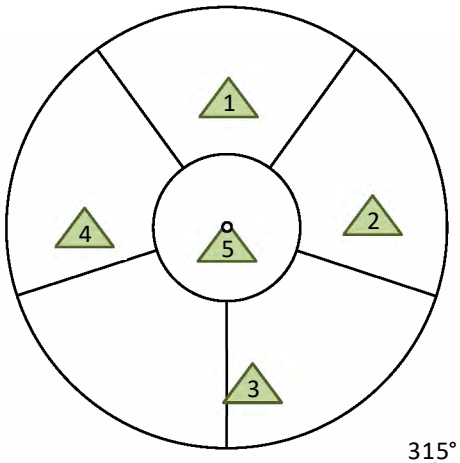
## Inspection Drawings

1) Vessel UT Layout

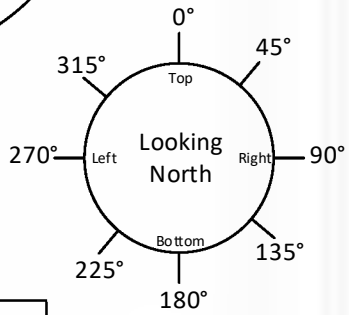



North Head




South Head




| CML | Description | Reading |
|---|---|---|
| 1 | South Head 12 O'Clock | 0.518 |
| 2 | South Head 3 O'Clock | 0.526 |
| 3 | South Head 6 O'Clock | 0.526 |
| 4 | South Head 9 O'Clock | 0.515 |
| 5 | South Head Center | 0.52 |
| 26 | North Head 12 O'Clock | 0.516 |
| 27 | North Head 3 O'Clock | 0.521 |
| 28 | North Head 6 O'Clock | 0.503 |
| 29 | North Head 9 O'Clock | 0.512 |
| 30 | North Head Center | 0.507 |

*Note:*
*0° - Top of Vessel*
*Reading should be taken in clockwise fashion starting at South end*

Legend

# Condition Monitoring Location (CML)

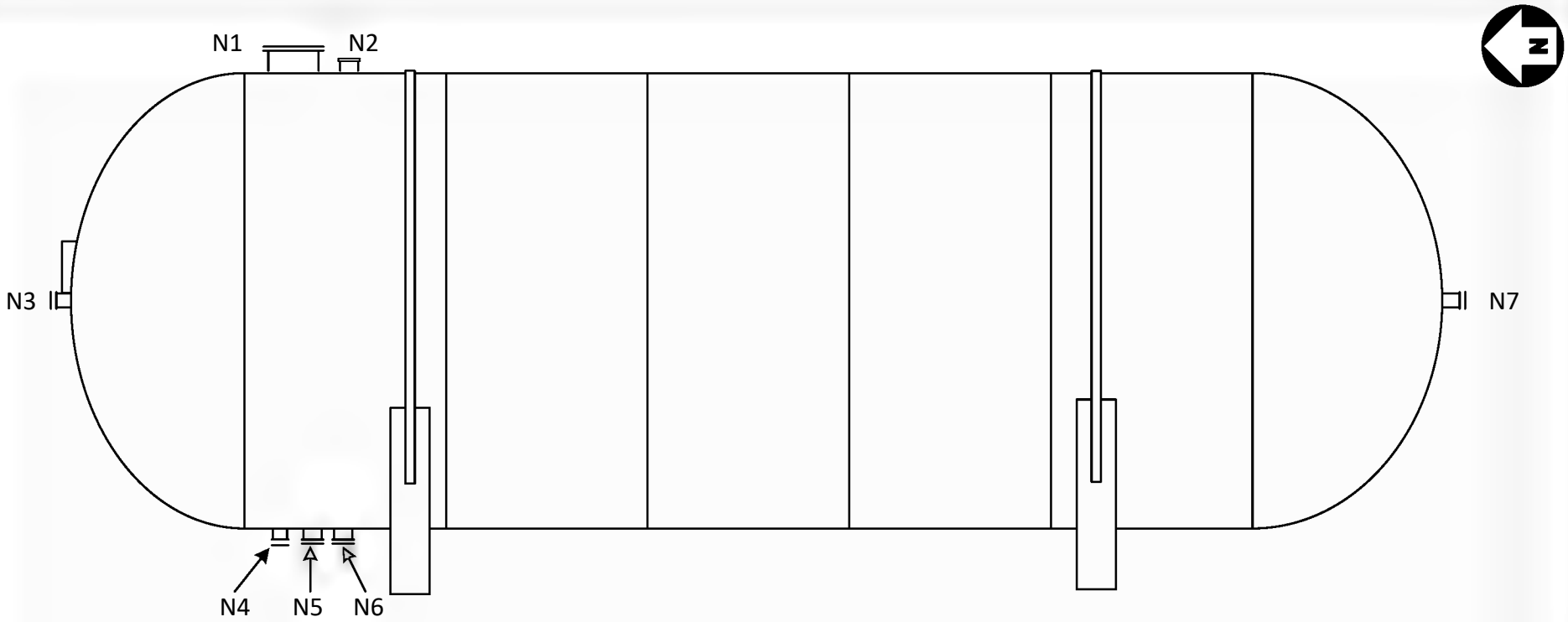
| | | | | |
|---|---|---|---|---|
| Instrument Serial No. | | | Inspection Date | |
| Transducer Serial No. | | | Inspection Type | Thickness Readings |
| Calibration Block | | | Company | |
| Flat or Curved | | Serial No. | Technician | |
| Material | | Manufacturer | | |
| Type | | | | |

**SAChem**

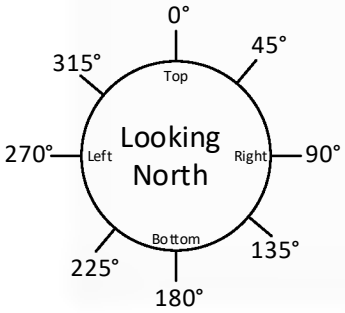
54-11-001 Trimethylamine Storage

CML Layout East \ West Head

Scale: None




Note:
Readings are taken in a clockwise fashion starting at 0°, north or top of nozzle.




Readings

| CML | Description | Size (in) | tmin* | 0° or Single | 90° | 180° | 270° |
|---|---|---|---|---|---|---|---|
| N1 | Manway | 16 | 1.440 | 0.578 | 0.577 | 0.575 | 0.578 |
| N2 | Vapor Out | 2 | 0.880 | 0.163 | 0.164 | 0.162 | 0.163 |
| N3 | Thermowell | 0.75 | 0.154 | 0.168 | 0.166 | 0.163 | 0.164 |
| N4 | Sight Gauge | 2.25 | 1.440 | 0.817 | 0.854 | 0.856 | 0.854 |
| N5 | Liquid In | 3 | 1.120 | 0.308 | 0.3 | 0.306 | 0.301 |
| N6 | Liquid Out | 3 | 1.120 | 0.306 | 0.308 | 0.308 | 0.302 |
| N7 | Gauge | 1 | 0.250 | 0.137 | 0.133 | 0.133 | 0.134 |
| | | | | | | | |
| | | | | | | | |

*Based on current ASME standards

Legend

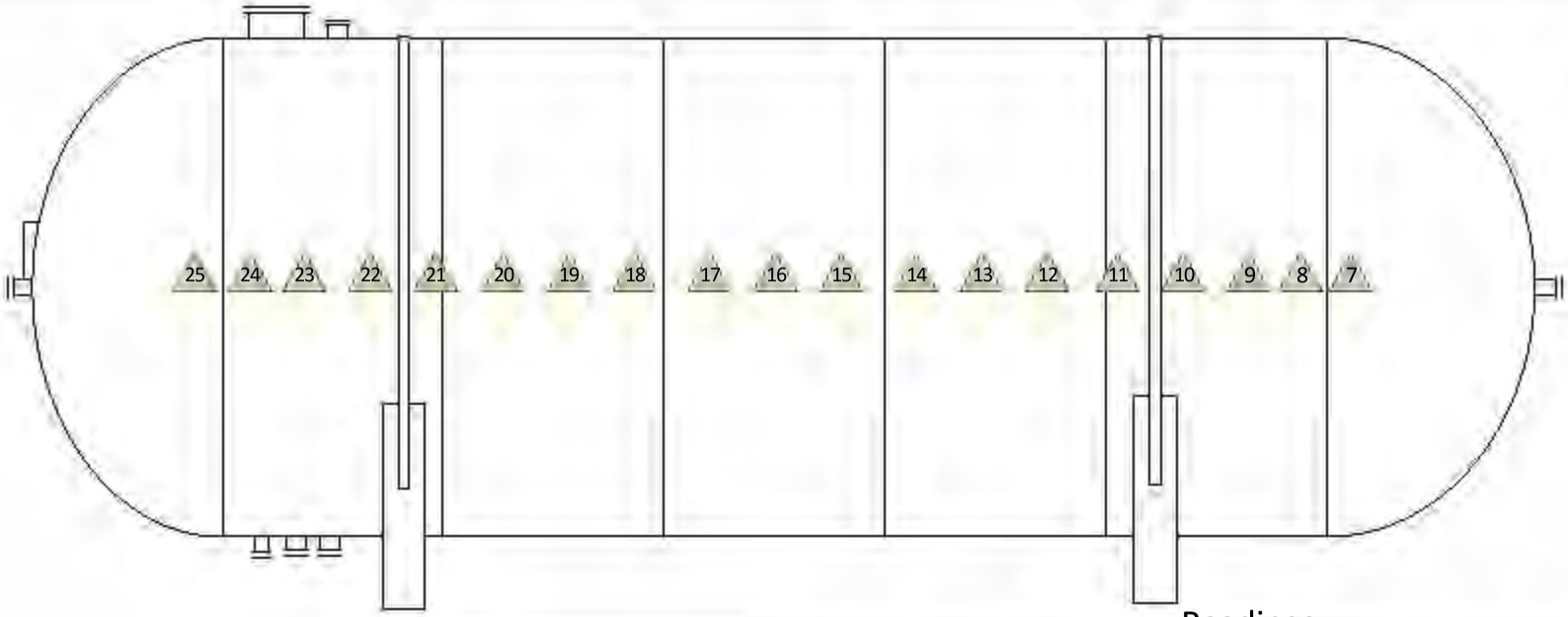
| | | | | |
|---|---|---|---|---|
| Instrument Serial No. | | | Inspection Date | |
| Transducer Serial No. | | | Inspection Type | Thickness Readings |
| Calibration Block | | | Company | |
| Flat or Curved | | Serial No. | Technician | |
| Material | | Manufacturer | | |
| Type | | | | |

**SAChem**

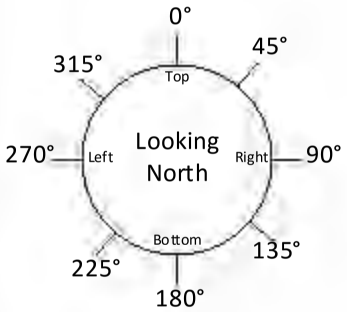
54-11-001 Trimethylamine Storage

Nozzle Readings

Scale: None




*Note:*
*0°- Top of Vessel*
*Reading should be taken in clockwise fashion starting at South end*




## Readings

| CML | Description | 0° or Single | 45° | 90° | 135° | 180° | 224° | 270° | 315° |
|---|---|---|---|---|---|---|---|---|---|
| 7 | 2" from South Head seam -Head side | 0.507 | 0.501 | 0.498 | 0.493 | 0.51 | 0.499 | 0.494 | 0.494 |
| 8 | 2" from South Head seam -Shell side | 0.81 | 0.809 | 0.813 | 0.813 | 0.812 | 0.811 | 0.811 | 0.809 |
| 9 | 2' | 0.811 | 0.814 | 0.807 | 0.815 | 0.813 | 0.813 | 0.81 | 0.812 |
| 10 | 4' | 0.813 | 0.812 | 0.809 | 0.813 | 0.811 | 0.815 | 0.81 | 0.814 |
| 11 | 6' | 0.811 | 0.81 | 0.813 | 0.811 | 0.809 | 0.811 | 0.811 | 0.81 |
| 12 | 8' | 0.807 | 0.814 | 0.812 | 0.818 | 0.819 | 0.817 | 0.813 | 0.814 |
| 13 | 10' | 0.82 | 0.818 | 0.819 | 0.825 | 0.822 | 0.824 | 0.817 | 0.815 |
| 14 | 12' | 0.819 | 0.822 | 0.82 | 0.825 | 0.817 | 0.822 | 0.818 | 0.819 |
| 15 | 14' | 0.816 | 0.818 | 0.816 | 0.817 | 0.821 | 0.814 | 0.814 | 0.814 |
| 16 | 16' | 0.811 | 0.814 | 0.811 | 0.804 | 0.82 | 0.824 | 0.818 | 0.819 |
| 17 | 18' | 0.811 | 0.811 | 0.813 | 0.808 | 0.817 | 0.826 | 0.822 | 0.821 |
| 18 | 20' | 0.804 | 0.808 | 0.802 | 0.797 | 0.808 | 0.823 | 0.814 | 0.804 |
| 19 | 22' | 0.807 | 0.811 | 0.81 | 0.801 | 0.808 | 0.809 | 0.809 | 0.807 |
| 20 | 24' | 0.807 | 0.809 | 0.804 | 0.801 | 0.807 | 0.809 | 0.809 | 0.81 |
| 21 | 26' | 0.806 | 0.804 | 0.808 | 0.797 | 0.799 | 0.801 | 0.805 | 0.804 |
| 22 | 28' | 0.807 | 0.806 | 0.809 | 0.806 | 0.813 | 0.814 | 0.81 | 0.811 |
| 23 | 30' | 0.81 | 0.808 | 0.804 | 0.806 | 0.818 | 0.812 | 0.811 | 0.811 |
| 24 | 2" from North Head seam- Shell side | 0.806 | 0.807 | 0.804 | 0.801 | 0.81 | 0.812 | 0.808 | 0.803 |
| 25 | 2" from North Head seam- Head side | 0.51 | 0.501 | 0.497 | 0.499 | 0.509 | 0.503 | 0.5 | 0.498 |

Legend

# Condition Monitoring Location (CML)

| | | | |
|---|---|---|---|
| Instrument Serial No. | R149470 | Inspection Date | |
| | | Inspection Type | Thickness Readings |
| | | Company | OILPRO |
| Flat or Curved | Serial No. 120509309 | Technician | J KILLINGSWORTH |
| Material | Manufacturer OLYMPUS | | J Killingsworth |

**SAChem**

54-11-001 Trimethylamine Storage

SHELL CAN Jerry Hartfield

Scale: None



rev 09/08/2016

# APPENDIX D

## Inspection Checklist

1) API 510 Checklist
2) In Lieu Of Guide Requirements

| API-510 PRESSURE VESSEL INSPECTION CHECKLIST | | | | Inspector: | Steve Sherman |
|---|---|---|---|---|---|
| Company: | SACHEM | | | Cert No.: | |
| Vessel: | 54-11-001 | Report No.: | 54-11-001-11 | Date: | 6/20/2017 |

## 1 FOUNDATION

### 1.1 Steel Members

| | | |
|---|---|---|
| a. | X | Visually inspect for pitting, and corrosion. |
| b. | X | Check fnd bolts secure with minimum thrd engagement |
| c. | X | Check for coating failures |
| d. | X | Check attachment welds for cracking and corrosion. |

### 1.2 Containment

| | | |
|---|---|---|
| a. | X | Inspect the area for buildup of trash, vegetation and obstructions. |
| b. | X | Inspect sump drain operation. |
| c. | X | Check that runoff rainwater drains away from the tank. |
| d. | X | Describe type of construction - Earthen, Concrete, Asphalt, Gravel |
| e. | | Inspect condition of containment. |

### 1.3 Concrete Foundation Supports

| | | |
|---|---|---|
| a. | X | Inspect for broken concrete, spalling and cracks. |
| b. | X | Inspect for erosion under foundation. |
| c. | X | Check for settlement around perimeter of tank. |

### 1.4 Wooden Saddle Support

| | | |
|---|---|---|
| a. | | Check for degradaded members (split, broken, dry rotted et.). |
| b. | | Inspect for errosion and vegetation tank fnd. |
| c. | | Check for settlement around perimeter of tank. |

## 2 SHELLS

### 2.1 External Visual Inspection

| | | |
|---|---|---|
| a. | X | Visually inspect shell surface for paint failures, pitting, corrosion, denting, out-of-round and part deformation. |
| b. | X | Clean angles and other components that form catch basins and check for degradation and corrosion.. |
| c. | X | Inspect nozzle penetration surfaces and welds for corrosion, cracking and deformation. |
| d. | X | Check for broken, unused insulation rod supports causing corrosion nodes. |
| e. | X | Visually inspect weld joints for cracking, pitting, corrosion and signs of leaking (product residue). |
| f. | X | Perform dye penetrant or magnetic particle tests if leaks or cracks are suspected. |
| g. | X | Check for proper grounding |

### 2.2 Internal Visual Inspection

| | | |
|---|---|---|
| a. | | Check atmospheric conditions, fill out and post safe entry permit form. |
| b. | | Appropriate and wear required PPE for safe entry. |
| c. | X | Inspect shell surfaces for coating failures, pitting and corrosion. |
| d. | | Inspect baffle plate surfaces &weld attachments for cracking, pitting, or corrosion. Check bolts are secure with minimum thread engagement. |
| e. | | Inspect heating coils surfaces and weld attachments, for cracking, pitting, corrosion. Check bolts are secure with minimum thread engagement. |
| f. | | Inspect pressure containing weld joints for cracking, pitting, and corrosion. |
| g. | X | Perform dye penetrant or magnetic particle tests if cracks are suspected. |

## 3 SHELL APPURTENANCES

### 3.1 Manways and Nozzles

| | | |
|---|---|---|
| a. | X | Inspect for cracks or signs of leakage on weld joints at nozzles, manways, and reinforcing plates. |
| b. | X | Inspect for shell plate dimpling around nozzles, caused by excessive pipe deflection. |
| c. | X | Inspect for flange leaks and leaks around bolting. |
| d. | X | Check flange bolts are secure and have minimum thread engagement. |
| e. | X | Inspect sealing of insulation around manways and nozzles. |
| f. | X | Check for inadequate manway flange and cover thickness on mixer manways. |
| g. | X | Check exposed flange and cover faces. |

### 3.2 Relief Devices

| | | | | | | |
|---|---|---|---|---|---|---|
| a. | X | Inspect for flange leaks and leaks around bolting. | | | | |
| b. | X | Check flange bolts are secure and have minimum thread engagement. | | | | |
| c. | | Record inlet and outlet sizes sizes. | Inlet | | Outlet | |
| d. | | Check that relief system outlet discharges to safe location (outside of building). | | | | |
| e. | | Record certification date | | | | |
| f. | | Record pressure setting | | | | |
| g. | | Record type and ID | | | | |

### 3.3 Shell-Mounted Sample Station

| | | |
|---|---|---|
| a. | | Inspect sample lines for function of valves and plugging of lines, including drain or return-to-tank line. |
| b. | | Check circulation pump for leaks and operating problems. |
| c. | | Test bracing and supports of sample system and equipment. |

| API-510 PRESSURE VESSEL INSPECTION CHECKLIST | | | | Inspector: | Steve Sherman |
|---|---|---|---|---|---|
| Company: | SACHEM | | | Cert No.: | |
| Vessel: | 54-11-001 | Report No.: | 54-11-001-11 | Date: | 6/20/2017 |

**3.4 Heater (Steam Coils)**

| | | |
|---|---|---|
| a. | | Inspect condensate drain for presence of oil, indicating leakage. |

**3.5 Mixer/Agitator**

| | | |
|---|---|---|
| a. | | Inspect for proper mounting flange and support. |
| b. | | Inspect for leakage. |
| c. | | Inspect condition of power lines and connections to mixer. |

**3.6 Deck Plate and Grating**

| | | |
|---|---|---|
| a. | X | Inspect deck plate for corrosion-caused thinning or holes (not drain holes) and paint failure. |
| b. | X | Inspect plate-to-frame weld for rust scale buildup. |
| c. | X | Inspect grating for corrosion-caused thinning of bars and failure of welds. |
| d. | X | Check grating tie down clips. Where grating has been retrofitted to replace plate, measure the rise of the step below and above the grating surface and with other risers on the stairway. |

**3.7 Stairway Stringers/Rungs/Treads**

| | | |
|---|---|---|
| a. | X | Inspect stairway stringers, rungs and treads for corrosion, paint failure and weld failure. |
| b. | X | Inspect stairway supports to shell welds and reinforcing pads. |
| c. | X | Inspect steel support attachment to concrete base for corrosion. |

**4 VESSEL JACKET**

**4.1 Jacket Shell**

| | | |
|---|---|---|
| a. | X | Identify type of jacket (half pipe, cylindrical, dimpled and spot welded, etc.) |
| b. | X | Measure and record pitch distances. |
| c. | X | Visually inspect shell surface for paint failures, pitting, corrosion, denting, out-of-round and part deformation. |
| d. | X | Clean angle support rings and inspect for corrosion and thinning on plate, annular space and welds. |
| e. | X | Inspect the shell-to-foundation seal or barrier. |
| f. | | Check for broken, unused insulation rod supports causing corrosion nodes. |
| g. | X | Visually inspect for weld joints for cracking, pitting, corrosion and signs of leaking (product residue). |
| h. | X | Perform dye penetrant or magnetic particle test if leaks or cracks are suspected. |

**5 INSULATION**

**5.1 Visual Inspection**

| | | |
|---|---|---|
| a. | X | Check for holes, missing portions, deterioration due to corrosion or abuse. |
| b. | X | Check for sufficient sealing, especially around vessel appurtenances. |
| c. | X | Check for wetness |

**6 PRESSURE /TEMPERATURE INDICATORS**

**6.1 Physical and Operating Conditions**

| | | |
|---|---|---|
| a. | X | Check that indicators are securely attached and operating properly. |
| b. | X | Check that indicators are in accessible locations and readable. |
| c. | X | Check that indicators have current calibration date |
| d. | X | Check for any damage, and corrosion build up. |

**7 ASME/NAME PLATE DATA**

**7.1 Physical Condition**

| | | |
|---|---|---|
| a. | X | Check that ASME plate is securely attached |
| b. | X | Check that ASME plate is in accessible location and readable. |

| API-510 PRESSURE VESSEL INSPECTION CHECKLIST | | | | Inspector: | Steve Sherman |
|---|---|---|---|---|---|
| Company: | SACHEM | | | Cert No.: | |
| Vessel: | 54-11-001 | Report No.: | 54-11-001-11 | Date: | 6/20/2017 |

**7.2 Record Following ASME Data**

| | | | |
|---|---|---|---|
| a. | X | Board Number | 51109 |
| b. | X | Serial Number | 140435–01 –6 |
| c. | X | Radiography Examination | R2 |
| d. | X | MAWP | 225 PSI @ 125 f |
| e. | X | MDMT | -20 |
| f. | X | Nominal Shell Thickness | 0.8125" |
| g. | X | Nominal Head Thickness | 0.53" |
| h. | X | Nominal Hieght | 130.25 |
| i. | X | Head Material | CS-A612 |
| j. | X | Shell Material | CS-A612 |
| k. | | Jacket Material | |
| l. | X | Manufacturer | Beaird |
| m. | X | Year Built | 1977 |

**8 VESSEL LAYOUT DRAWINGS**

**8.1 Record Following Data**

| | | |
|---|---|---|
| a. | X | Head and skirt lengths and type. |
| b. | X | Shell lines of support (Head Tang, Braces, Jckt Closures) |
| c. | X | Foundation support member dimensions and orientation |
| d. | X | Inside/Outside diameter. |
| e. | X | Vessel part nominal thicknesses. |
| f. | X | Nozzle layouts, sizes and uses. |

**9 UT THICKNESS READINGS**

**9.1 Measure and Record Vessel Part Thicknesses**

| | | |
|---|---|---|
| a. | X | Heads - (4) locations, (1) in each quadrant. |
| b. | X | Shell - (4) locations, (1) in each quadrant for each course. |
| c. | X | Jacket - As required (at least (4) locations). |
| d. | X | Nozzles - (4) locations, (1) in each quadrant for 2 |

**Notes:**

This inspection exceeds all recommend a criteria for NDE above.